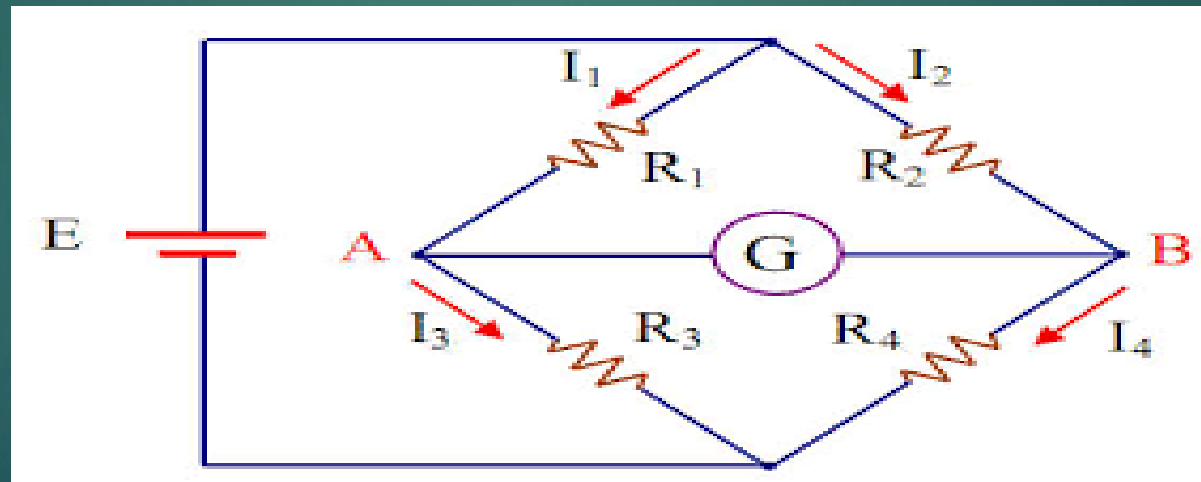


Experiment #(12)

Wheatstone Bridge

جسر ويتستون



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Objective

- ❑ To construct a slide wire Wheatstone bridge.
- ❑ To measure the value of an unknown resistance.

Apparatus

Power Supply



Slide Wire



Galvanometer



Banana Wires



Resistor



Resistance Box



Theory

Wheatstone Bridge: An instrument or a circuit consisting of four resistors or their equivalent in series, used to determine the value of an unknown resistance when the other three resistances are known.

The Wheatstone bridge network consists of four resistors R_1 , R_2 , R_3 , and R_4 interconnected as shown in the illustration Fig.1 to form the bridge.

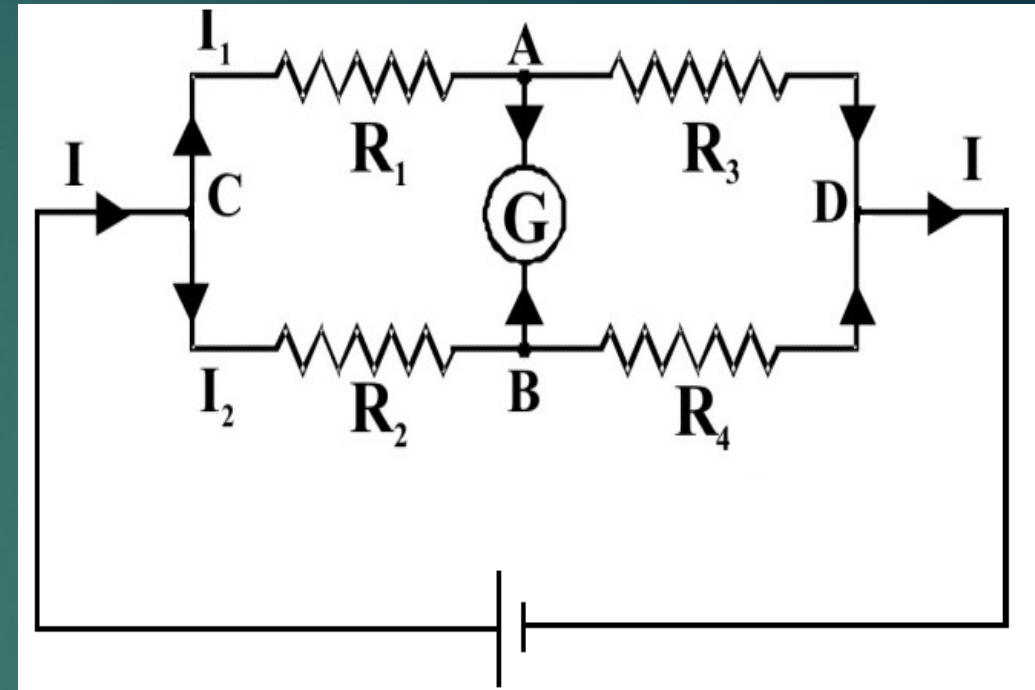


Figure 1

The Galvanometer current will be zero and this adjustment will be independent of the supply voltage. Thus, when the bridge is balanced ($I_G=0$) this implies $V_A=V_B$, then

$$V_{AC} = V_{BC}$$

$$I_1 R_1 = I_2 R_2 \dots \dots \dots (1)$$

$$\text{and, } V_{AD} = V_{BD}$$

$$I_1 R_3 = I_2 R_4 \dots \dots \dots (2)$$

deviding(1)by(2)

$$\frac{R_1}{R_3} = \frac{R_2}{R_4} \dots \dots \dots (3)$$

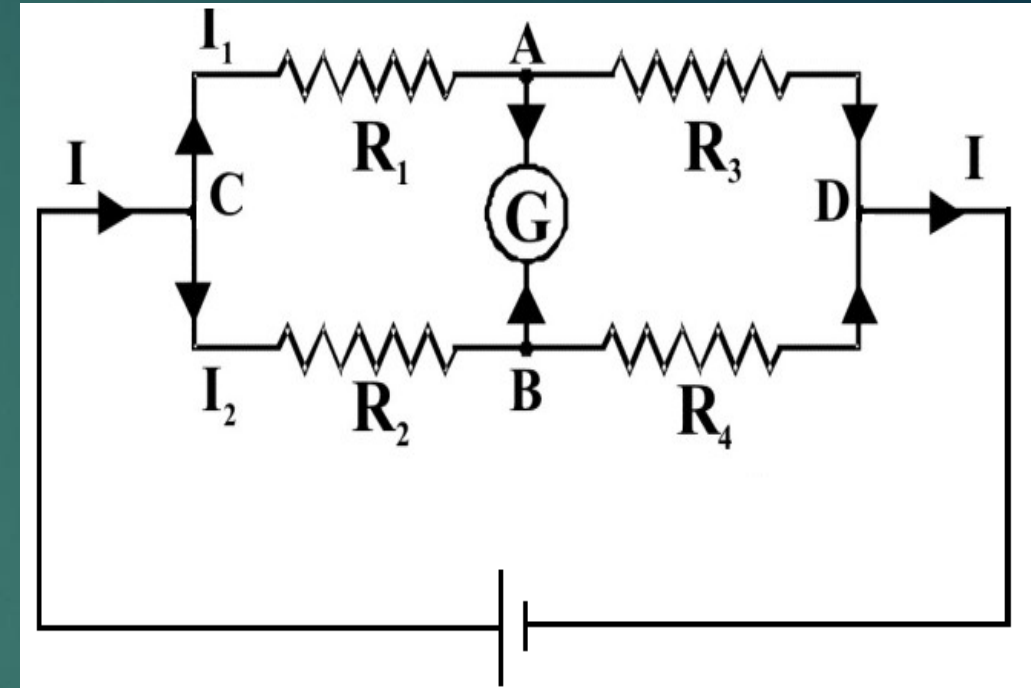


Figure 1

In the following circuit Fig.2 If $I_G = 0$, then

$$\frac{R_x}{R} = \frac{R_b}{R_a} = \frac{\left(\frac{\rho b}{A}\right)}{\left(\frac{\rho a}{A}\right)} = \frac{b}{a}$$

$$R = R_x \frac{a}{b} \dots\dots\dots (4)$$

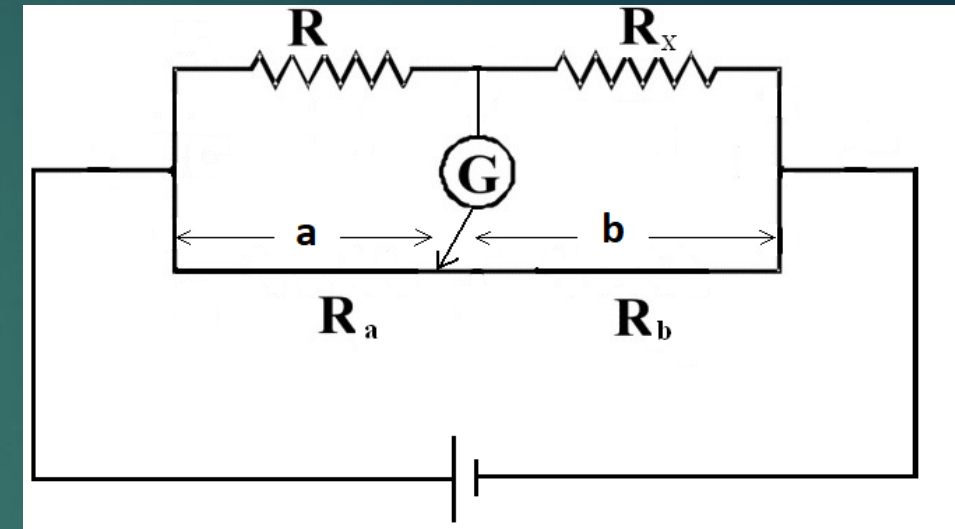


Figure 2

Procedure

1. Connect the circuit shown in Figure 2. Have your lab instructor check it before you make the connection to the power supply.
2. Change the value of the resistance box R and vary the position of the sliding tap slightly to balance at any desired point along the length of the slide wire.
3. Record the values of the lengths a and b in table .1.
4. Draw the relation between R and the ratio a/b and calculate the slope.
5. From the slope calculate the unknown resistance R_x .

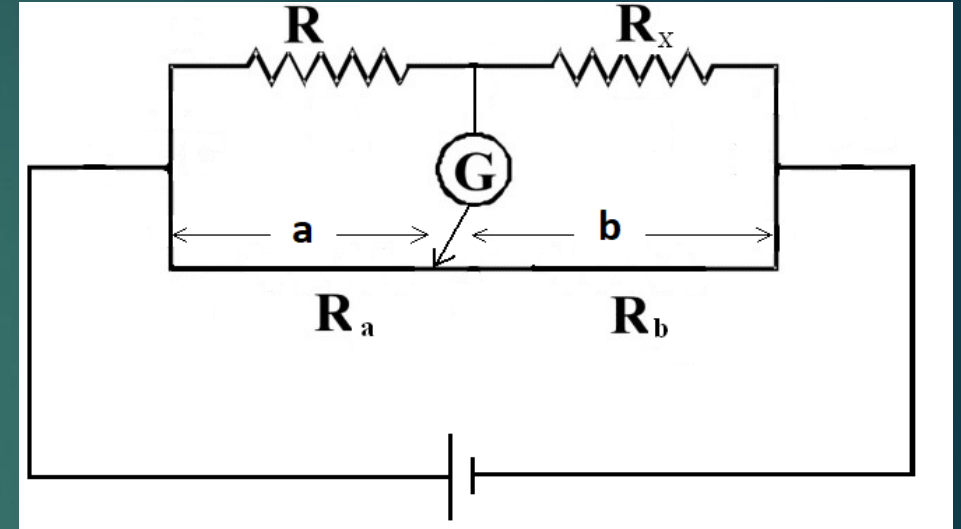


Figure 2

Table 1(R_x)

trial	$R(\)$	$a(\)$	$b(\)$	a/b
1				
2				
3				
4				
5				



THANK YOU

